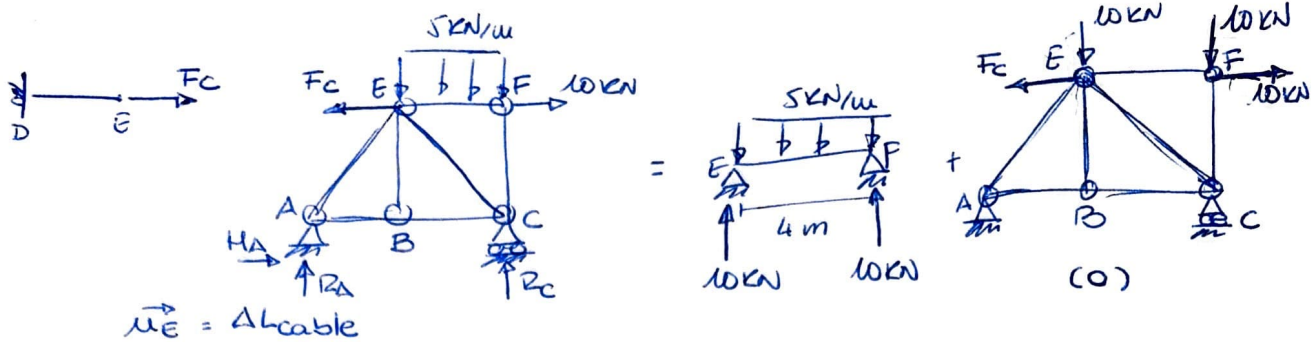
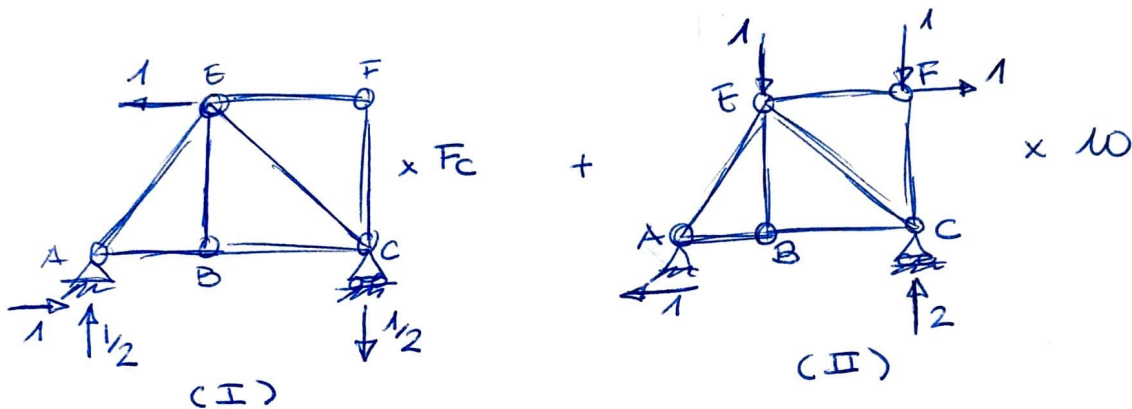


EXERCÍCIO 1



1.)

Superposição de estados: (0) = (I) + (II)



Barra	L_i	N_i^I	N_i^{II}	$(N_i^I)^2 L_i$	$N_i^I N_i^{II} L_i$	N_i^0	N_i^{III}	$N_i^0 N_i^{III} L_i$
AB	4	$-1/2$	1	1	-2	8'987	1	35'948
BC	4	$-1/2$	1	1	-2	8'987	1	35'948
AE	$4\sqrt{2}$	$-\sqrt{2}/2$	0	$2\sqrt{2}$	0	-1'432	0	0
BE	4	0	0	0	0	0	0	0
CE	$4\sqrt{2}$	$\sqrt{2}/2$	$-\sqrt{2}$	$2\sqrt{2}$	$-4\sqrt{2}$	-12'709	0	0
EF	4	0	1	0	0	10	0	0
CF	4	0	-1	0	0	-10	0	0
				Σ	$[2+4\sqrt{2} \cdot -4-4\sqrt{2}]$			71'896

+ tração
- compressão

$$\vec{u}_E = \frac{1}{EA_{\text{barras}}} \left[\sum (N_i^0)^2 L_i F_c + \sum N_i^I N_i^{III} L_i \cdot 10 \right]$$

$$N_i^0 = N_i^I \cdot F_c + N_i^{III} \cdot 10$$

$$\Delta L_{\text{cable}} = \frac{F_c \cdot 4}{EA_{\text{cable}}}$$

$$\vec{u}_E = \Delta L_{\text{cable}} \rightarrow -\frac{1}{10^5} \left[(2+4\sqrt{2}) F_c - (4+4\sqrt{2}) 10 \right] = \frac{F_c \cdot 4}{10^4}$$

$$|F_c = 2'026 \text{ kN}|$$

2) REACCIONES

$$\vec{H}_A = 1 \cdot F_c - 10 = -7'974 \text{ kN}$$

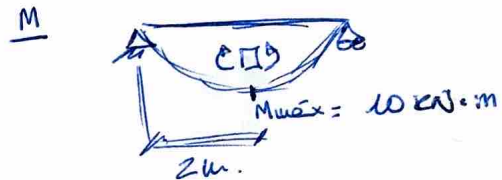
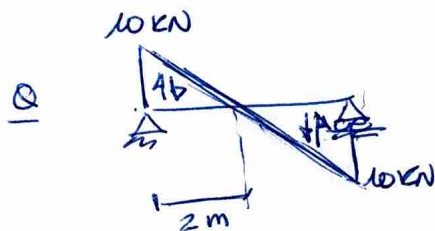
$$\vec{R}_A = \frac{1}{2} \cdot F_c = 1'013 \text{ kN}$$

$$\vec{R}_C = -\frac{1}{2} F_c + 20 = 18'987 \text{ kN}$$

3) ESFUERZOS EN TODAS LAS BARRAS DE LA ESTRUCTURA.

Esfuerzos axiales $\rightarrow$ En la Tabla $\rightarrow N_i^0$

Q y M $\rightarrow$ Solo barra EF

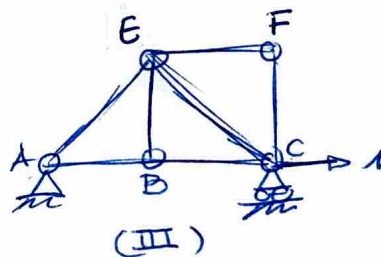


4) DESPLAZAMIENTOS NUDO C

$$\vec{v}_C = 0$$

$$\vec{u}_C = \sum N_i^0 N_i^{\text{III}} \cdot \frac{L_i}{EA_b}$$

N_i^{III} en la tabla



$$\vec{u}_C = \frac{1}{10^5} \cdot 71'896 = 7'19 \cdot 10^{-4} \text{ m.}$$

5) MÍNIMO ΔT EN BARRA CE PARA QUE EL CABLE NO TRABAJE.

$$\begin{cases} \vec{u}_E = 0 \\ F_c = 0 \end{cases} \Delta T_{ce} ?$$

$$\text{Si } F_c = 0 \rightarrow N_c^0 = N_c^{\text{II}} \cdot 10$$

$$\vec{u}_E = \sum N_i^0 (-N_i^{\text{I}}) \cdot \frac{L_i}{EA_b} + (-N_{CE}^{\text{I}}) \cdot \alpha \Delta T \cdot L_{CE} = 0$$

$$0 = -\frac{1}{EA_b} [-(4+4\sqrt{2}) \cdot 10] - \frac{\sqrt{2}}{2} \cdot 10^{-5} \Delta T \cdot 4\sqrt{2}$$

$$\boxed{\Delta T = 24'14^\circ \text{C}}$$